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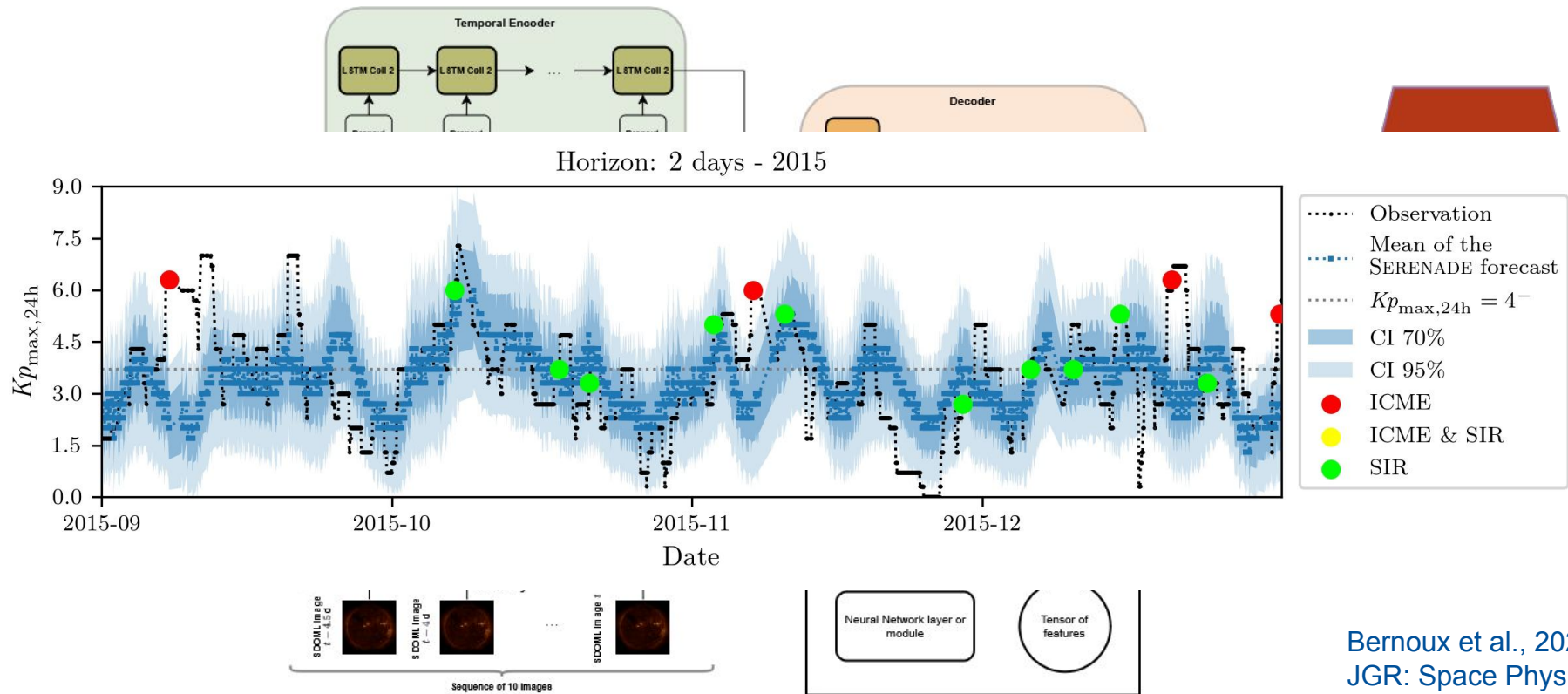
Improved geoeffectiveness prediction via DL and Extended Solar EUV Imaging Dataset

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ESWW 2025

Context: the SERENADE model

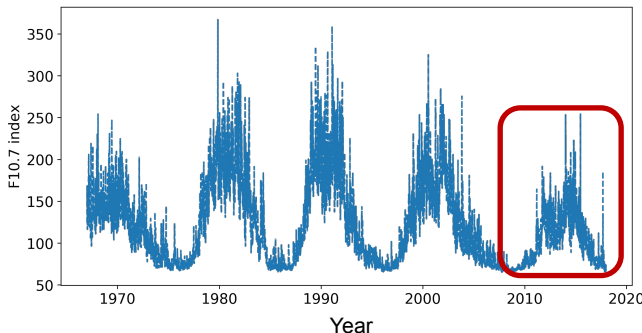
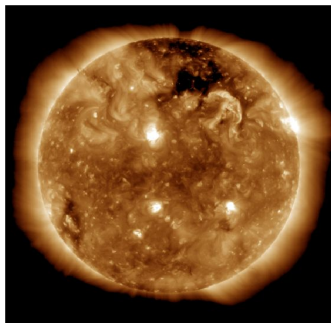


Bernoux et al., 2022
JGR: Space Physics

The “SDOML” EUV dataset

□ ML-ready SDO/AIA images (Galvez et al., 2019)

- EUV imaging
- ✓ Reduced size: 512 x 512 pixels
- ✓ Calibrated, corrected
- ✓ Corrupted images removed
- ✓ 6 min cadence
- ✓ ~10 TB



- Two versions:
 - V1 : **2010 – 2018**, .npz files, hosted by JSOC
 - V2 : **2010 – 2020**, Zarr datastore, hosted in an AWS S3 bucket
- Used by many studies for forecasting, detection of structures, segmentation, super-resolution, etc. □ 110 citations currently (Google Scholar)
- Issues :
 - Only covers **SC24**, which was weak
 - Missing last 5 years and all the events that occurred in this period
 - Original FITS metadata discarded

The questions

- How much of an **improvement** could be brought to SERENADE (and by extension, probably to other models) **by simply adding the past 5 years of data** to the pipeline?
- In other words, *to what extent did training models using only data from the weak solar cycle 24 resulted in underperforming models?*
- Can we expect ML/DL models trained with data from solar cycle 25 to be better *simply because there is more data that is more representative?*
- What happens if we **lose access to SDO/AIA** data? Could we simply **switch instruments**/sources? What would be the **loss of accuracy**?

What we are doing today

- 1) **Extend and create** SDO/AIA and SOHO/EIT **ML-prepared datasets**
- 2) **Retrain SERENADE** using combinations and subsets of the datasets
- 3) **Evaluate** the models and compare

Recreating and extending an ML-prepared dataset for... SDO/AIA

Data source:

- Images are retrieved from the **MEDOC SDO/AIA Level-1 archive**
- Queries include date range, wavelength, and cadence (**6 min**) and **keep only the best-quality** (QUALITY=0) observations.

Processing:

1. Verification: each downloaded L1 FITS file is checked for completeness, valid dimensions, no NaNs/Inf, and correct header keywords. **Invalid files are excluded.**
2. Pointing correction using the latest pointing table (JSOC).
3. Centering & rotation & rescaling to a common solar radius of 0.76 arcsec

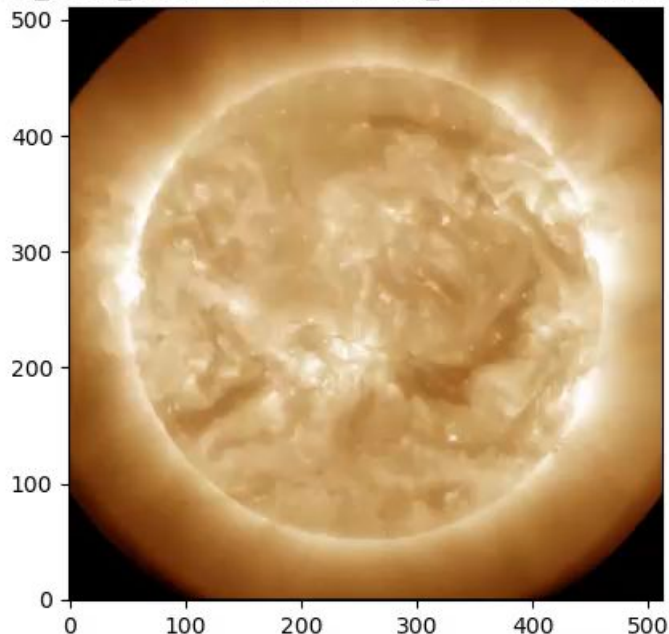
- Basically **L1.5 processing + few extra steps**
- Equivalent to what is done in the SDOML paper
- Done using **Sunpy and aiapy** for most steps

4. Downsampling from $4096 \times 4096 \rightarrow 512 \times 512$

5. Instrument degradation correction with the JSOC correction table.

Recreating and extending an ML-prepared dataset for... SDO/AIA

aia.MLP_193A_2024-01-01T00-00-40_339821736.image_MLP.fits



- 4 EUV channels (171, 193, 211, 304)
- 05/2010 – 05/2025
- ~5M images
- ~5 TB
- Images kept in FITS files
- Metadata kept and updated

(Re)creating an ML-prepared dataset for... SOHO/EIT

Data source:

- Images are retrieved from the **ESA SSA TAP** service (**SOHO/EIT Level-1** calibrated products).
- Queries are limited to a specific wavelength, date range, and **science objectives** (“CME WATCH” / “FULL SUN”).

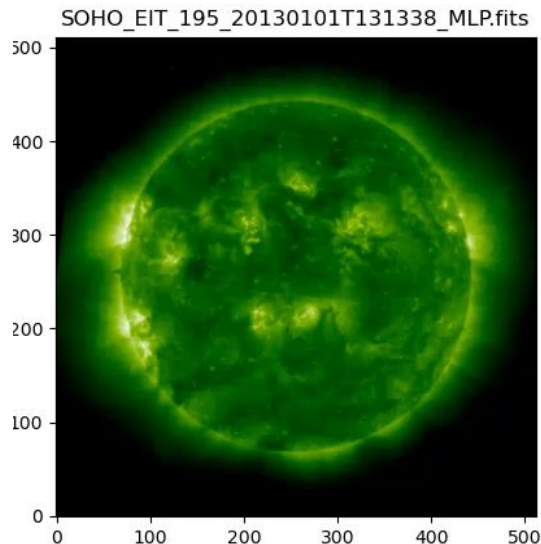
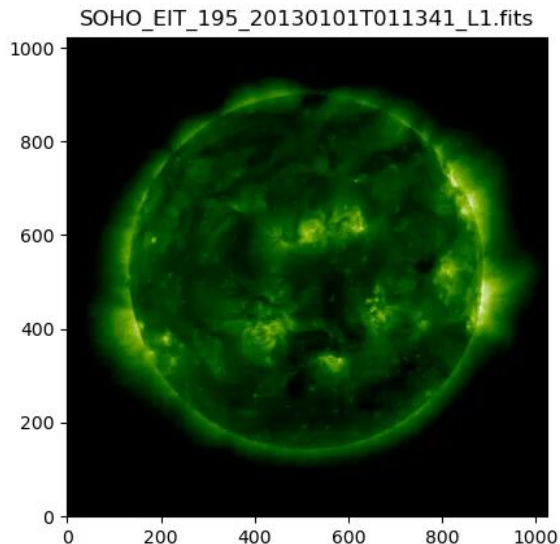
Processing:

1. Confirm presence of image data, absence of NaN/Inf, correct dimensions (512×512 or 1024×1024), and clean header keywords. **Bad data are excluded.**
2. Rotation & rescaling to a common solar radius of 976 arcsec.
3. Centering and filling empty “outside” regions with a low-percentile off-limb value.

- Some of the SDO/AIA L1.5 steps are already performed for SOHO/EIT L1
- No “quality” flag equivalent to SDO/AIA
- Done using **Sunpy** for most steps

4. Down-sample 1024×1024 images to 512×512 (super-pixel averaging); 512×512 images are left

Recreating and extending an ML-prepared dataset for... SOHO/EIT

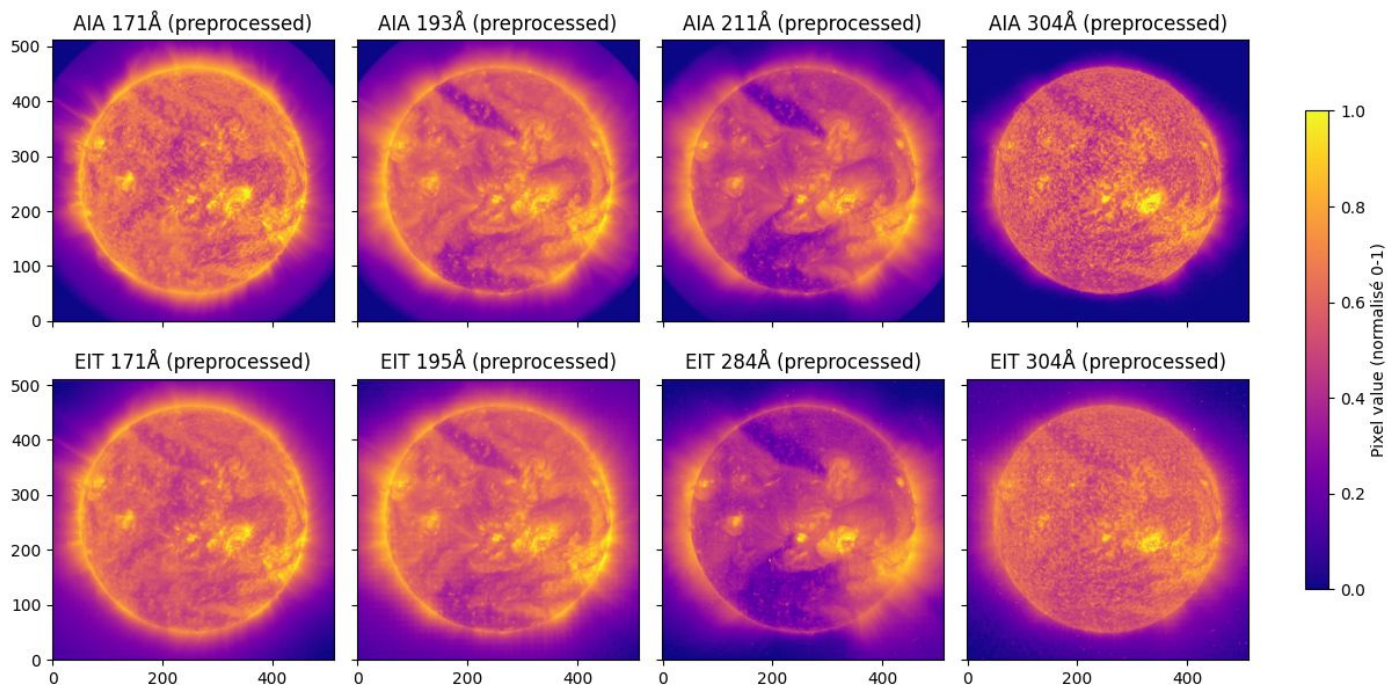


- 4 EUV channels (171, 195, 284, 304)
- 1996 – 2024
- ~500K images
- ~0.5 TB
- Images kept in FITS files
- Metadata kept and updated
- Would require human intervention to be fully cleaned

Additional preprocessing steps

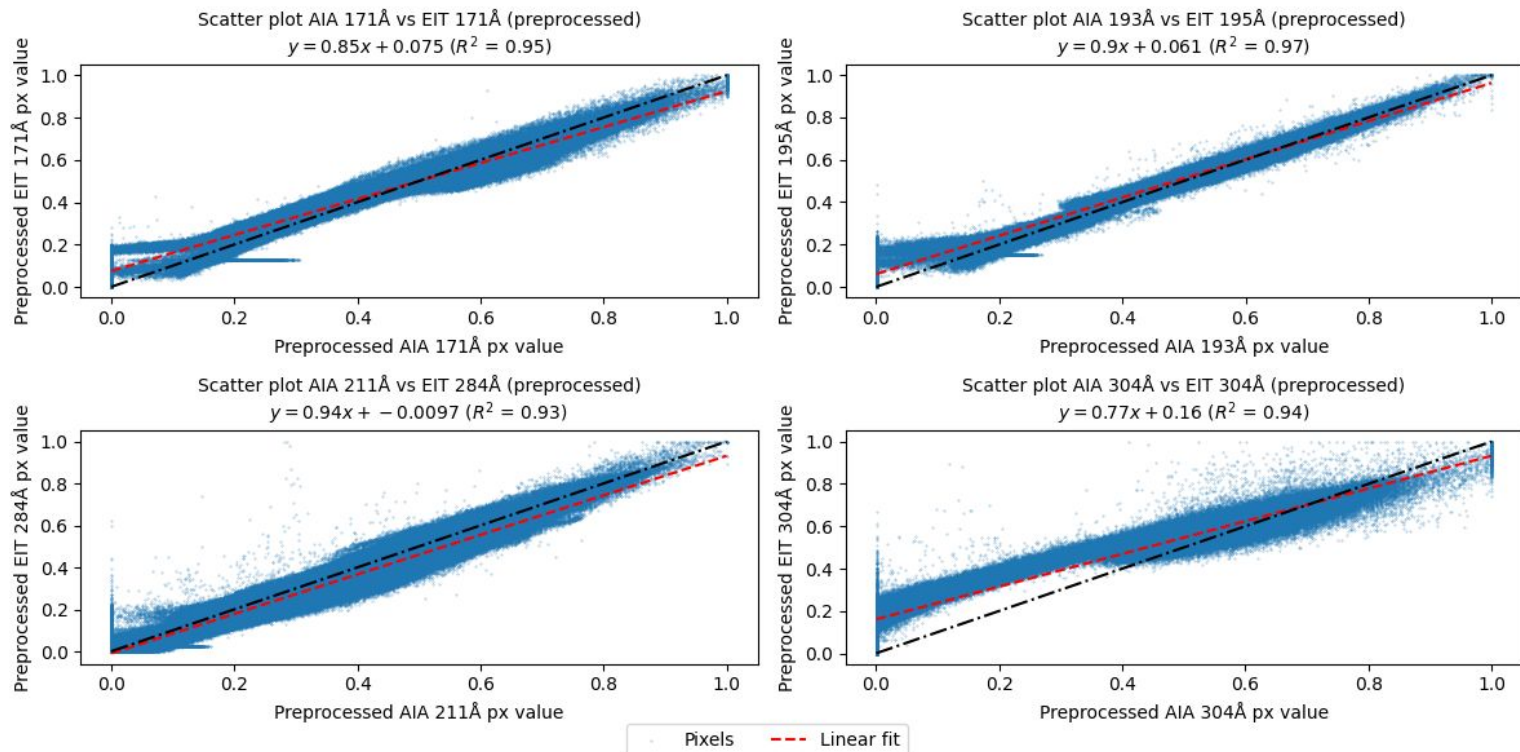
- Here we only use 193A or 195A images with SERENADE
- All images are:
 - Log10-scaled
 - Clipped using 1st and 99.9th percentiles
 - Rescaled to 0 – 1 range
 - Turned into “fake RGB” for compatibility with pretrained feature extractor by copying the single channel

Example AIA-MLP and EIT-MLP images



Examples for images taken around 2015-03-15 01:00:00

Example AIA-MLP and EIT-MLP images



Examples for images taken around 2015-03-15 01:00:00

Details about the SERENADE pipeline

Model's architecture (used in this study)

- Inputs :
 - 10 images separated by 12hr
- Feature extractor:
 - ~~GoogLeNet~~ (dim: 1024) ☐ **DINOv3-base** (dim: 768)
- Temporal Encoder (LSTM):
 - Hidden size: ~~2048~~ ☐ 256
 - Layers: ~~2~~ ☐ 4
 - Attention: ~~None~~ ☐ Multihead attention on top
 - Normalization: ~~None~~ ☐ LayerNorm
- Decoder:
 - MLP 256 ☐ 64 ☐ 16 ☐ 1 & 1
- Output:
 - Kp (max, 24hr) or **Solar wind bulk velocity** (from OMNI 1h)
 - Normal distribution ☐ **LogNormal distribution** (i.e. the model predicts the mean and the standard deviation of the distribution)
- Loss function:
 - Variance-regularized Negative Log-Likelihood
$$\mathbf{L}(\mu, \sigma, \mathbf{y}) = \frac{1}{2} \log(\sigma^2) + \frac{(y - \mu)^2}{2\sigma^2} + \lambda \frac{1}{\sigma^2 + \varepsilon}$$
- Validation : 2001, 2011
- Test : **2006**, 2014, **2017**, 2020
- Train : remaining years

Details about the SERENADE pipeline: model evaluation

Regression metrics

Root Mean Square Error (lower is better) between the observation and the median forecast:

$$\textbf{RMSE}(M, Y) = \sqrt{(\mu_i - y_i)^2}_{i=1 \dots N}$$

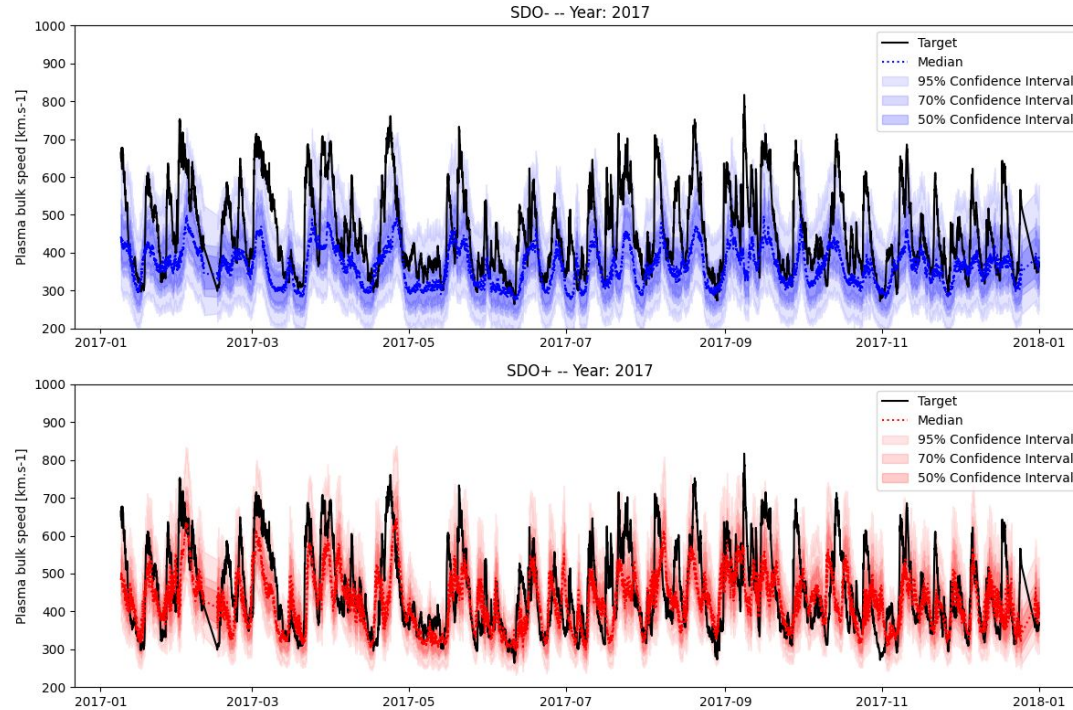
Continuous Ranked Probability Score (lower is better):

$$\textbf{CRPS}(M, S, Y) = \int_{-\infty}^{+\infty} (F_{\mu, \sigma}^i(x) - \mathbf{1}(x - y_i))^2 dx \quad i=1 \dots N$$

→ Generalization of Mean Absolute error to probabilistic forecasts

Results for the SDO/AIA extension

3-days ahead forecasts

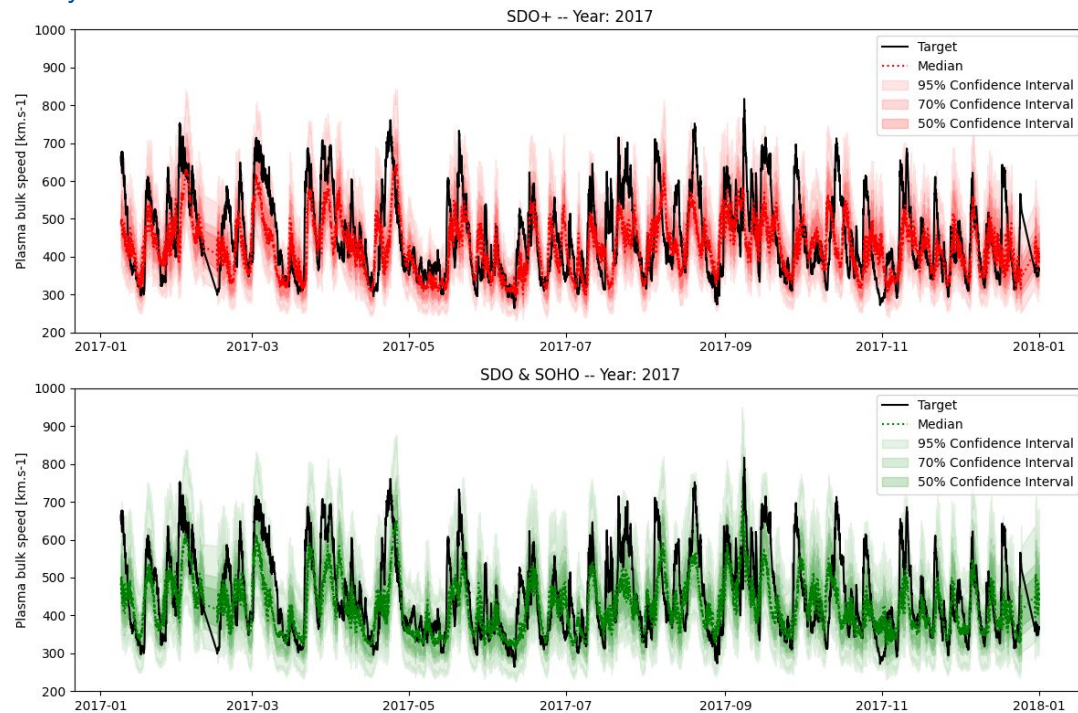


Metrics for 2017	SDO-	SDO+
RMSE [km.s-1]	124	95
CRPS [km.s-1]	70	53

- ✓ Much better results with the extended dataset!
- ❖ Both models can actually be optimized but SDO+ is always better

Results for the SDO + SOHO extension

3-days ahead forecasts

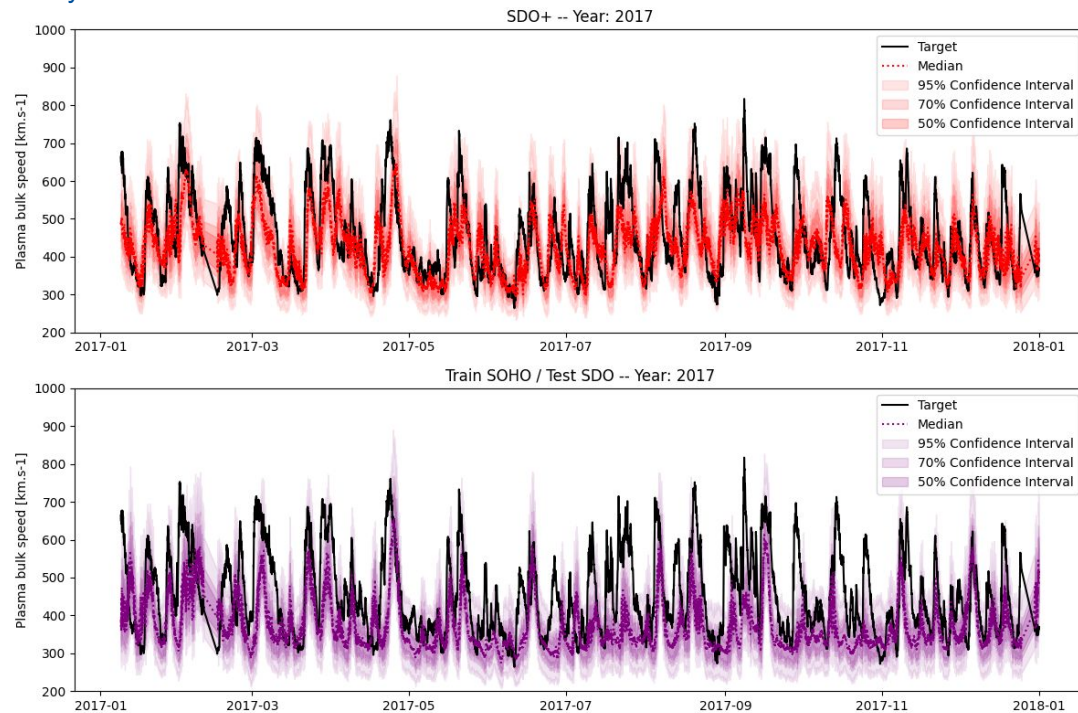


Metrics for 2017	SDO+	SDO & SOHO
RMSE [km.s ⁻¹]	95	89
CRPS [km.s ⁻¹]	53	48

- ✓ Again better results when adding SOHO to the training set
- ❖ Visually not very different though: accurate and adequate validation is important

Training with SOHO/EIT... and using with SDO/AIA

3-days ahead forecasts

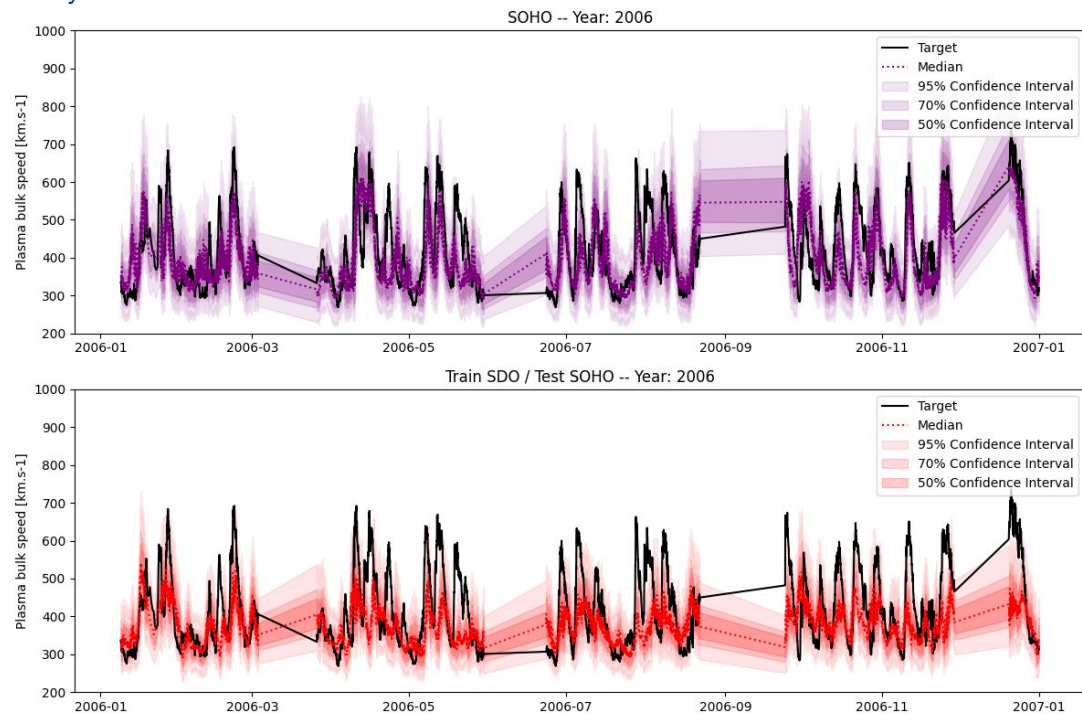


Metrics for 2017	SDO+	Train SOHO / Test SDO
RMSE [km.s-1]	95	127
CRPS [km.s-1]	53	75

- ✓ The model is still able to provide some results...
- ✓ ... but they are not as accurate
- ❖ This means that despite the images being very similar, the model is sensitive to instrument-specific “noise” and less stable

Training with SDO/AIA... and using with SOHO/EIT

3-days ahead forecasts



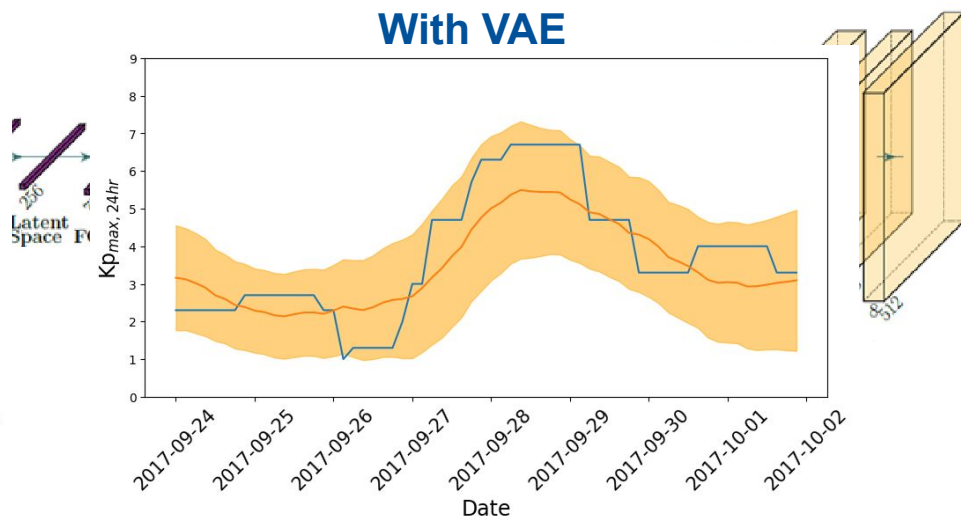
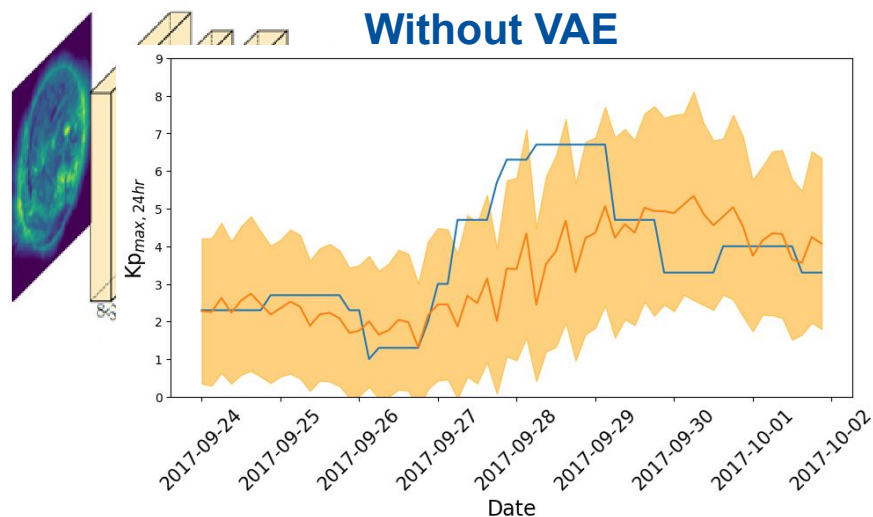
Metrics for 2006	SOHO	Train SDO+ / Test SOHO
RMSE [km.s ⁻¹]	82	100
CRPS [km.s ⁻¹]	45	55

- ✓ Still degraded results...
- ✓ ...even though the difference is not as large (SDO+ train set is large)
- ❖ Results are not “stretched” enough as if the model was lacking “confidence”

Some conclusions and messages

- There are probably some DL models published these past few years that would benefit from being (re)trained on data from SCs 23 and 25
- More data = more good... but:
 - Need **maintained** datasets OR **processing script**, provided by **people who know the data**
 - That remains “user-friendly” and enables research and proof-of-concept developments
 - Need **funding** to create and **maintain** these datasets and/or scripts
- DL models may not be as robust as wished to new data sources
 - This **needs to be taken into consideration** when deploying DL models operationally
 - **Dedicated strategies** (pretraining, few shot learning, etc.) could (should?) be applied to robustify models to diverse/changing data sources
 - More in-depth studies are needed to **quantify** this behavior with other datasets and models

- Repeating this study using a **solar-imaging dedicated feature extraction model** such as the **Variational AutoEncoder presented in Tahtouh et al. 2025** (JGR: MLC)



- Should increase pure forecasting “performance” (Tahtouh+ used SDO- dataset, we can now do better)
- Might help improve robustness and stability when facing new data due to a better organized latent space
- Not done here because training the VAE required 1 to 2 weeks per experiment (using a single GPU)



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Thank you for your attention

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